

Nexum Suum – Surge Tank & Deaerator (DA) Troubleshooting SOP

■ Page 1: Overflow, Venting, Oxygen Removal, Feedwater Surge & Level Control

If DA or Surge Tank Overflows:

- Check Modulating Valve for Feedwater Control – May be stuck open.
- Inspect Level Transmitter or Float Switch for Drift or Debris.
- Verify Pump Starts on Level Demand, Not Continuous.

If Oxygen Content in Feedwater is Too High:

- Inspect Steam Sparger or Tray Section for Fouling or Erosion.
- Check DA Steam Pressure – Should match rated design.
- Review Operating Temp – Target $\geq 212^{\circ}\text{F}$ for full deaeration.

If Steam is Venting Excessively:

- Check Steam Regulator or Trap Failure Causing Overpressure.
- Confirm Vent is Not Plugged or Discharging Condensate.

If Surge Tank Level Fluctuates Rapidly:

- Check for Pump Short Cycling.
- Ensure Inlet Makeup Valve Has Proper Orifice and Delay Time.
- Evaluate Return Line Air Entrapment or Flash Steam.

Nexum Suum – Surge Tank & Deaerator SOP (Cont'd)

■ Page2: Symptom-to-Action Mapping+Real-World Thresholds

Symptom	Action
Tank Overflow	Check level control valve, transmitter drift, float switch.
High Oxygen in Feedwater	Review DA pressure/temp, inspect sparger/tray performance.
Excessive Steam Venting	Inspect trap, regulator, vent restriction or condensate block.
Rapid Level Cycling	Slow fill valve response or short-cycling pump logic.

■ Operating Thresholds:

- DA Tank Operating Temp: $\geq 212^{\circ}\text{F}$ (ideal $225\text{--}230^{\circ}\text{F}$)
- Steam Pressure to DA: 5–15 psi typical
- Oxygen Content: < 7 ppb (parts per billion) for feedwater
- Tank Vent Rate: Minimal visible plume; adjust for lowest O₂ carryover
- Level Control: ± 2 " setpoint band around midpoint
- Surge Tank Fill Time: < 2 minutes per fill cycle
- Conductivity (Feedwater): $< 50\ \mu\text{S}/\text{cm}$ to prevent corrosion
- Steam Trap Discharge Temp: $200\text{--}212^{\circ}\text{F}$ for effective flash recovery